

Sheet 4 - information econ

(a)

	π_1	π_2	π_3
e_0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
e_1	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$$u(w, e) = \sqrt{w} - e.$$

The principal solves the problem

$$\min_{w_1, w_2, w_3} E[w] = \frac{1}{4} w_1 + \frac{1}{2} w_2 + \frac{1}{4} w_3$$

$$\text{s.t. IR: } \frac{1}{4} \sqrt{w_1} + \frac{1}{2} \sqrt{w_2} + \frac{1}{4} \sqrt{w_3} - 1 \geq 0$$

$$\text{IC: } \frac{1}{4} \sqrt{w_1} + \frac{1}{2} \sqrt{w_2} + \frac{1}{4} \sqrt{w_3} - 1 \geq \frac{1}{3} (\sqrt{w_1} + \sqrt{w_2} + \sqrt{w_3})$$

Since we're told $e=1$ is optimal.

- We know that IR will bind - otherwise, the principal could reduce payments in all states s.t. IC still holds, but pay lower E wages.
- Also, we know IC will bind - otherwise, there would be some ~~other~~ point between (w_1, w_2, w_3) and $(10, 10, 10)$ where IR is satisfied, IC still holds, but E wage costs for principal are lower, by risk-aversion / concavity of $\sqrt{\cdot}$.

The Lagrangian is

$$\begin{aligned} \mathcal{L}(w_1, w_2, w_3; \lambda, \mu) = & \frac{1}{4} w_1 + \frac{1}{2} w_2 + \frac{1}{4} w_3 \\ & - \lambda \left(\frac{1}{4} \sqrt{w_1} + \frac{1}{2} \sqrt{w_2} + \frac{1}{4} \sqrt{w_3} - 1 \right) \\ & - \mu \left(-\frac{1}{12} \sqrt{w_1} + \frac{1}{6} \sqrt{w_2} - \frac{1}{12} \sqrt{w_3} - 1 \right) \end{aligned}$$

$$\text{with FOCs } \frac{\partial \mathcal{L}}{\partial w_1}: \frac{1}{4} - \lambda \cdot \frac{1}{8} w_1^{-1/2} + \mu \cdot \frac{1}{24} w_1^{-1/2} \stackrel{!}{=} 0$$

$$\frac{\partial \mathcal{L}}{\partial w_2}: \frac{1}{2} - \lambda \cdot \frac{1}{4} w_2^{-1/2} - \mu \cdot \frac{1}{12} w_2^{-1/2} \stackrel{!}{=} 0$$

$$\frac{\partial \mathcal{L}}{\partial w_3}: \frac{1}{4} - \lambda \cdot \frac{1}{8} w_3^{-1/2} + \mu \cdot \frac{1}{24} w_3^{-1/2} \stackrel{!}{=} 0$$

as well as IC and IR binding.

Q1a yes, correct!

Examining FOCs for w_1 and w_3 , we have

$$\frac{1}{\sqrt{w_1}} \cdot \left(\frac{\lambda}{9} - \frac{\mu}{24} \right) = \frac{1}{4} = \frac{1}{\sqrt{w_3}} \cdot \left(\frac{\lambda}{9} - \frac{\mu}{24} \right), \text{ which guarantees}$$

that $w_1^* = w_3^*$, since the bracketed term cannot be zero for the product to be $\frac{1}{4}$.

From IC, then $\frac{1}{2}(\sqrt{w_1} + \sqrt{w_2}) = 11$. And from IC,

$$\frac{1}{6}(\sqrt{w_2} - \sqrt{w_1}) = 1.$$

$$\text{So, } \sqrt{w_2} = 14 \Rightarrow w_2^* = 196; w_1^* = w_3^* = 64.$$

Given that KKT conditions are assumed to be necessary and sufficient, we have the optimal contract as above.

b): Note that the monotone likelihood ratio does not apply:

$$\frac{\Pr(\pi_1 | e_1)}{\Pr(\pi_1 | e_0)} < \frac{\Pr(\pi_2 | e_1)}{\Pr(\pi_2 | e_0)} > \frac{\Pr(\pi_3 | e_1)}{\Pr(\pi_3 | e_0)}$$

r_1 r_2 r_3 , with $r_1 = r_3$, (and this is why $w_1^* = w_3^*$, since both provide the same evidence to principal of effort).

Q1b (i) yes, correct!

So, since $w_2^* > w_3^*$, the agent might destroy revenue in π_3 -worlds to reach the π_2 -level, and earn higher wages.

ii. The principal needs to make sure they incentivise the agent to truthfully deliver all revenue, i.e. that the agent is as well-compensated in π_3 -worlds as π_2 -worlds. (Note that it seems possible that the principal could get higher net profits even while allowing destruction, if it keeps wage bill low, but we ignore that.) So we'd impose the constraint $w_1 \leq w_2 \leq w_3$.

Given that the unconstrained optimum has $w_3 > w_2$, we'd expect this monotonicity constraint to bind at the new optimum, since the problem is convex. The argument in (a) that $w_1 = w_3$ will no longer hold, since we need to introduce new multipliers for the constraints $w_2 - w_1 \geq 0$, $w_3 - w_2 \geq 0$ that break symmetry (and if

they did still hold, that would imply flat wages, which certainly fails IC). However, IC and IR remain unchanged and binding.

So substituting $w_2 = w_3$ into IR, $\frac{1}{4} \sqrt{w_1} + \frac{3}{4} \sqrt{w_2} = 11$, and IC, $\frac{1}{12} (\sqrt{w_2} - \sqrt{w_1}) = 1$

1b (ii) yes, correct, great reasoning

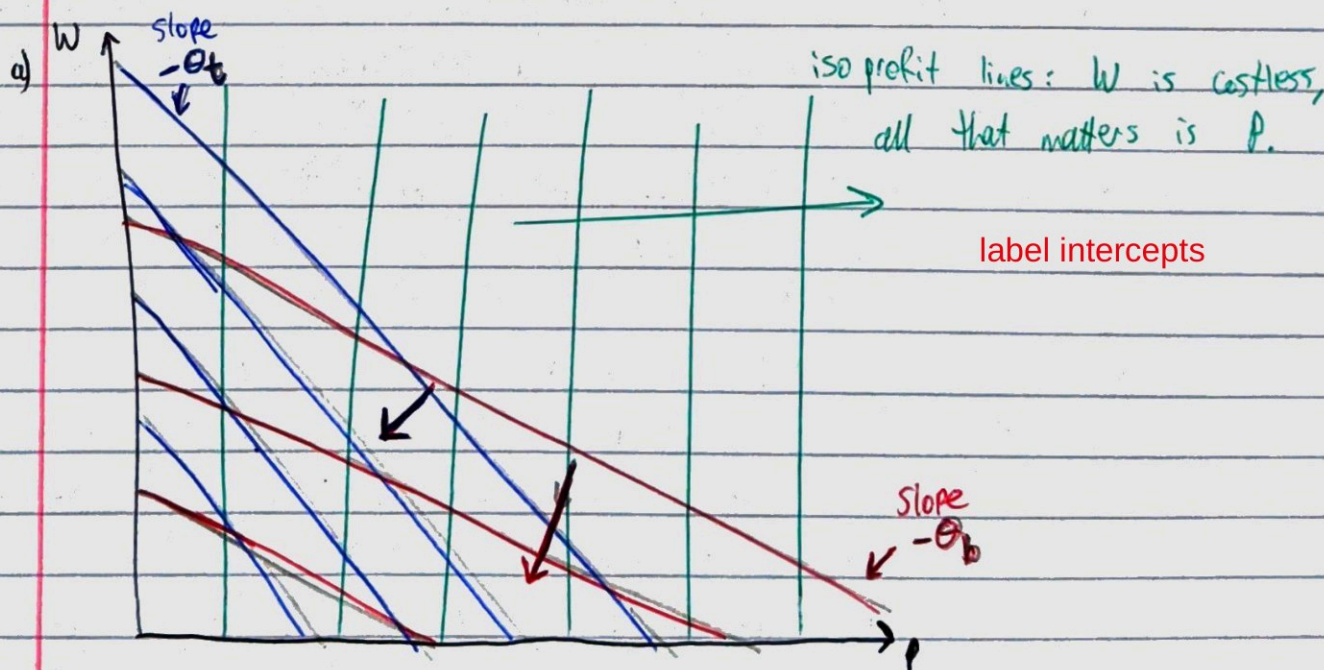
$\Rightarrow w_2 = 196$ as before, $w_1 = 4$, $w_3 = 196$.

c) In part (a), expected wages were $\frac{1}{2} \cdot 196 + \frac{1}{2} \cdot 64 = 130$. In part (b), they're $\frac{3}{4} \cdot 196 + \frac{1}{4} \cdot 4 = 148$.

yes, correct, could provide additional intuition on why agency costs must necessarily be higher in the second case (i.e., to prevent revenue destruction))

Agency cost is the reduction in principal's expected profit relative to the full-information (first-best) optimum. It's equal to the risk premium of the lottery produced by the wage structure for the agent. It must increase from part (a) to part (b), since we introduce a new constraint and thus the new optimum in strictly convex problem is worse. Note that in the first-best case, the agent simply receives her reservation utility from a flat amount when she puts in $e=1$, so agency costs are 9 and 27 respectively.

2. ~~2.1~~ 2.2: $W = v - \theta_b P - \bar{u}$ for each indiff. curve value $\bar{u}$



By revelation principle, WLOG we can consider direct, truth-telling-inducing mechanisms.
 $\therefore$ The business solves

$$\max_{\{(P_b, w_b), (P_t, w_t)\}} \lambda (P_t - c) + (1 - \lambda) (P_b - c)$$

$$\text{s.t. } P_t \geq 0, P_b \geq 0,$$

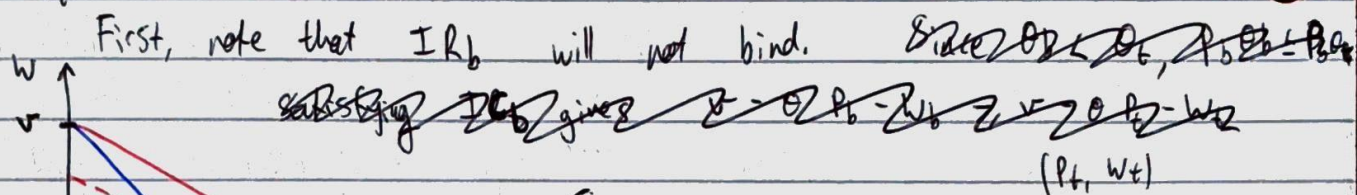
$$IC_t: v - \theta_t P_t - w_t \geq v - \theta_t P_b - w_b$$

$$IC_b: v - \theta_b P_b - w_b \geq v - \theta_b P_t - w_t$$

$$IR_t: v - \theta_t P_t - w_t \geq 0$$

$$IR_b: v - \theta_b P_b - w_b \geq 0$$

b) Suppose IR_t was slack. Then the airline could slightly increase P_t



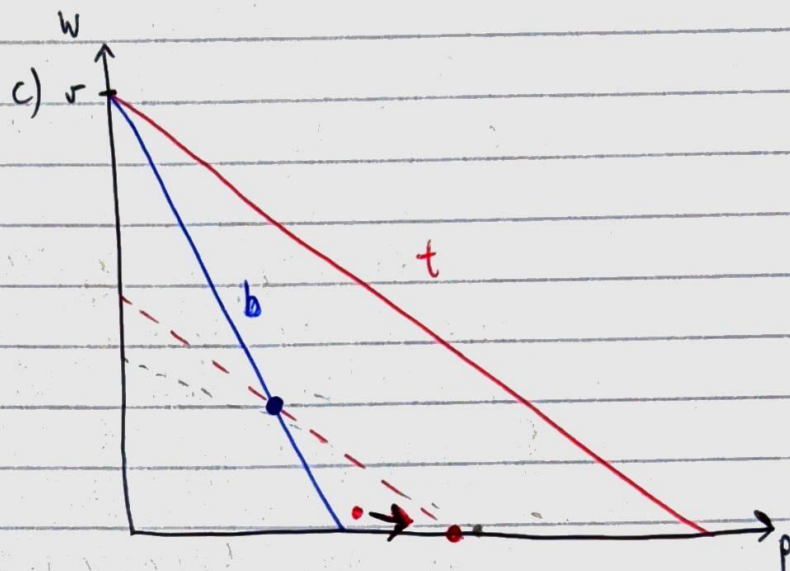
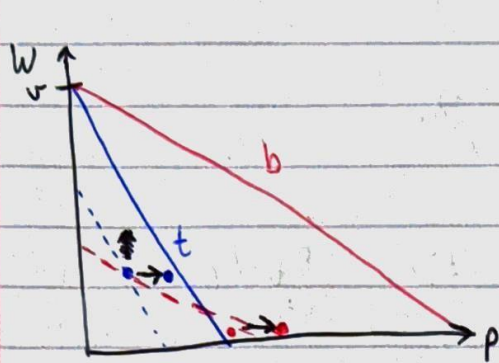
Suppose $\bullet$ is some contract satisfying IR_t . (It should actually be drawn below — since lower indifference curves are better, but argument still holds).

The indifference curve of b passing through $\bullet$ is at higher utility since they pay more for any given w . So for IC_b to hold, the contract $\bullet$ must offer at least as high utility. But then it's certainly $\geq$ reservation utility 0 given that $\bullet$ already gave b positive utility.

Algebraically: from IR_t , $v - w_t \geq \theta_t P_t$. And since $\theta_b \leq \theta_t$ and prices true, $\theta_b P_b < \theta_t P_b$.

$$\begin{aligned} \text{So } v - \theta_b P_b - w_b &\geq v - \theta_b P_t - w_t \quad \text{by } IC_b, \\ &\geq \theta_t P_t - \theta_b P_t \quad \text{by } IR_t \\ &\geq 0 \quad \Rightarrow IR_b \text{ must hold with slack.} \end{aligned}$$

This means IR_t must bind: if it didn't, firm could increase P_t and P_b by the same amount, earning more profit and keeping IR_t satisfied for a sufficiently small ϵ .



We know IC_b binds, so $\bullet (p_b, w_b)$ lies on — passing through $(0, v)$, i.e. indiff. curve with $\bar{u} = 0$.

If the business contract $\bullet$ was strictly preferred to $\bullet$ by b -types, the firm could increase p_b to get more profits ~~without~~ without affecting the constraints' satisfaction. And if $\bullet$ specifies $w_b > 0$, note that the firm can costlessly reduce wait time w_b while increasing p_b along the ind. curve $----$. So optimally $w_b = 0$ and $\bullet$ lies on ind. curve passing through $\bullet$.

d) We know IC_t and IC_b both bind, and $w_b = 0$.

$$IC_t: p_t = \frac{-w_t + v}{\theta_t} \quad IC_b: \theta_b(p_b - p_t) = w_t$$

$$\Rightarrow p_b = \frac{w_t}{\theta_b} + p_t = \frac{w_t}{\theta_b} + \frac{-w_t + v}{\theta_t}$$

$$\text{So profits} = \lambda \left(\frac{-w_t + v}{\theta_t} - c \right) + (1 - \lambda) \left(\frac{w_t}{\theta_b} + \frac{-w_t + v}{\theta_t} - c \right)$$

$$= \frac{w_t - v}{\theta_t} + (1 - \lambda) \frac{w_t}{\theta_b} - c$$

$$= w_t \left(-\frac{1}{\theta_t} + \frac{(1 - \lambda)}{\theta_b} \right) + \frac{v}{\theta_t} - c, \text{ where the only choice variable is } w_t.$$

Examining the coefficient on w_t , we have:

$0 < \theta_b < \theta_t$, so $\frac{1}{\theta_t} < \frac{1}{\theta_b}$. And since $\lambda \in [0, 1]$, the sign of the coefficient $\frac{1-\lambda}{\theta_b} - \frac{1}{\theta_t}$ depends on λ .

When $\frac{1-\lambda}{\theta_b} - \frac{1}{\theta_t} > 0 \Rightarrow \theta_b/\theta_t < 1-\lambda$, then increasing w_t increases profits. For I_t , we can't have $w_t > v$, so the firm offers tourists $w_t = v$ and $p_t = 0$, making profits on businesspeople.

i.e. When $\theta_b/\theta_t < 1-\lambda$ contracts are separate
 $t: (0, v)$ and $b: (v/\theta_b, 0)$ in (p, w) space

and when $\theta_b/\theta_t \geq 1-\lambda$ contracts for both are the same, ~~businesspeople~~ i.e. $(v/\theta_b, 0)$

Intuitively, when the two types are similar (so θ_b/θ_t is not small) or when there's only a small fraction of b-types, it's not worth it for the firm to push up w_t and cut prices for t-types s.t. b-types prefer their separate contract.

In separating eq*, all surplus goes to principal. In pooling eq*, the b-types get some information rents of $v(1/\theta_b - 1/\theta_t) > 0$

e) Note that the airline loses money on every t-type in separating eq* for any $c > 0$. So really they'd prefer to withdraw that contract entirely and serve only b-types.

- If $c > v/\theta_b$ then the firm should just shut down since marginal ^{excess revenue} costs ^{even} for b-types.
- If $v/\theta_t < c < v/\theta_b$ the firm should ~~implement a separating eq*~~ but only offer the b-contract, since they make negative profits on t-contracts. Pooling would lose money on each sale.
- The profit from exclusion of t-types is $(1-\lambda) \cdot (v/\theta_b - c)$, whereas separating eq* gives $(1-\lambda) v/\theta_b - c$, which is λc lower from transporting the tourists for free.

- With very low costs $c < v/\theta_b$, the firm could pool or serve businesspeople only. latter is better when

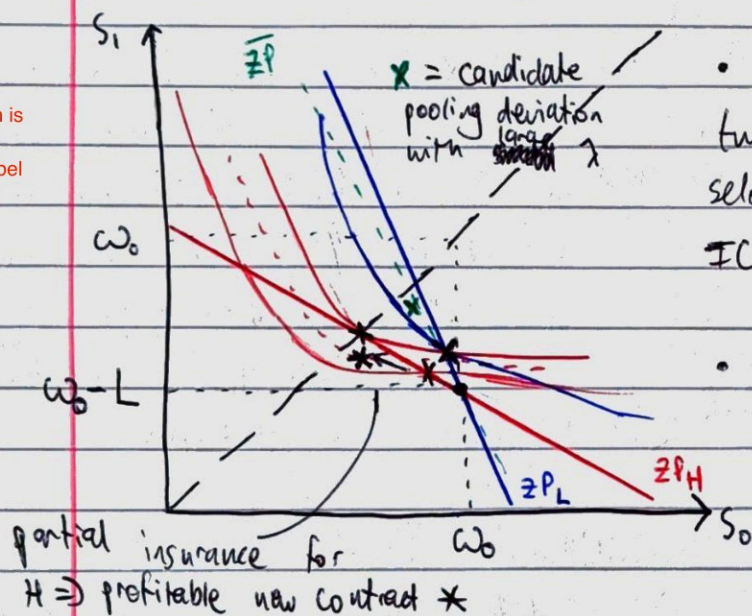
$$(1-\lambda)(v/\theta_b - c) > v/\theta_t - c$$

$$\Leftrightarrow v \left(\frac{1-\lambda}{\theta_b} - \frac{1}{\theta_t} \right) + \lambda c > 0$$

good

3a) A Rothschild-Stiglitz equilibrium satisfies:

- Every contract offered obtains zero expected profit, and
- No new contract could be introduced with positive expected profit (holding others fixed)



The diagram is correct, well done, but label eqbm better

- In the ^(candidate) separating eq^t, we have two contracts available and each type selects the one intended for them, i.e. IC_L, IC_H, PC_L, PC_H all hold.

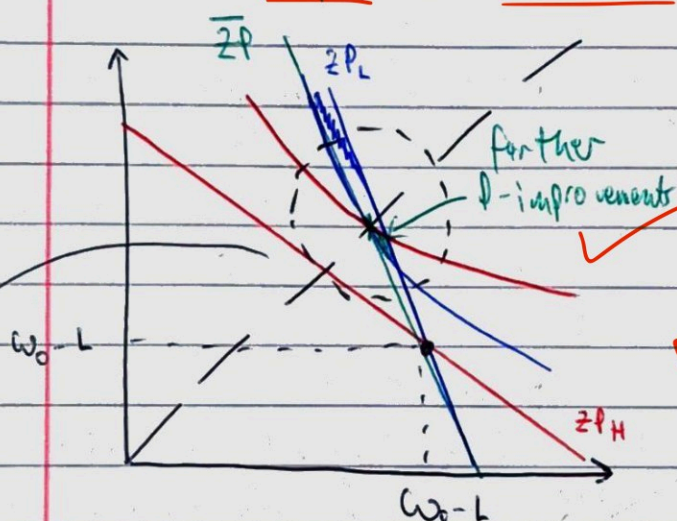
- Competition means each contract must be actuarially fair, i.e. lie on the zero-profit line for each type (otherwise the expected profit)

- But if less than full insurance were offered to high-risk types, then a profitable new contract would be to insure a slightly larger quantity at just above actuarially fair price.
- So, we can't have less than full insurance for high-types.

- For eq^t to exist, we need to have a sufficiently ~~large~~ small fraction of low-risk customers λ that a pooling deviation is not profitable. If λ is large, then a new contract could be introduced that both types prefer to separating ones, but still gets positive profits because population-wide risk is low enough.

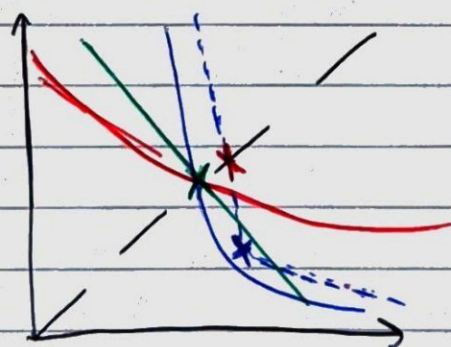
b) The social planner could offer a pooling contract like * described above. For sufficiently large λ , this is a offerable without a deficit (since L-types are numerous enough to cross-subsidise H), and as already argued is preferred by both types to their separating contracts.

In particular, she could offer one ~~any~~ contract of full insurance at actuarially-fair population-wide risk price. And there's no need to worry about a competitor entering and skimming off all the L-types with a cheaper but lower-coverage contract. Why not?



• At x , since the two types have different MRSs, it's actually possible to have further Pareto improvements: give a cheaper but lower-insurance contract to L-types, and a full-insurance, ~~more~~ more-subsidised contract to H-types.

The condition for Pareto-improvements is the reverse of eq^t existing in a) - we need enough L-types (large λ) to be able to cross-subsidise H-types.



using L-types to insure them more.

You can improve on pooling by this separation. H-types still get full insurance because otherwise we could increase their coverage (and EU) by

There's a tradeoff between budget balance (and needing to charge L-types more to cross-subsidise) and information revelation IC for the L-types.

c) To extract maximum surplus, the monopolist will only want to provide customers with reservation utility as far as possible. IR_H won't bind, since for the combination of IR_L and IC_H plus H-types always wanting more insurance at a given price ensures slackness.

• When λ is low, most of the profits would come from H-type consumers. The monopolist needs to ensure Hs don't use the L contract, if he's to serve both. But this would require offering ~~is~~ a very distorted (i.e. low-coverage) contract s.t. IR_L binds. (If he didn't, then could make both contracts more expensive). Hs do get full insurance since that's the efficient allocation of risk and is the way for monopolist to pay the least money in expectation for a given $\bar{u}_H$ s.t. IC_H binds. (Possibly even IR_L fails and it's best to exclude Ls.)

• When λ is large, ^{or excluding} distorting L's contract is costly because they would make up most of the monopolist's profits, so it's preferred ^{can't} for ~~him~~ to sell them a larger quantity. The monopolist ~~might~~ exclude H-types because any contract satisfying IR_L guarantees IR_H . So ~~then~~ he serves both at ~~+~~ one pooling contract of full insurance charged s.t. Ls are at reservation utility. Although monopolist loses money in expectation on every sale to Hs, they're sufficiently infrequent that this is worth it to sell a larger quantity of profitable coverage to L-types. ~~It~~

[or maybe separates with partial insurance to Ls?]

• So in either case, there's full coverage for H-types, and they get information rents, while Ls have zero surplus, or are even excluded.

No he does not pool. Pooling H with L will make monopolist lose on H types' profit. Pooling is suboptimal when offering a menu of contracts through cross subsidy can allow maximum profit for the monopolist